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N-body Simulations of Spiral Bending Waves in Planetary Rings

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ABSTRACT

Draft. This mock paper currently holds the derivation of the vertical epicyclic motion used by the `epi_int_3d` drift step (Section 2), the vertical gravity that one streamline exerts on another (Section 3), its relation to `epi_int_lite`'s in-plane streamline gravity (Appendix B), and a study of the streamline method's resolution limit (Appendix C), and the vertical restoring force of a bending wave (Appendix D), and the theory of spiral bending waves at a vertical resonance with its tests (Section 4): the forced response of massless particles, and the bending wave that the satellite launches in a self-gravitating ring.

1. INTRODUCTION

The `epi_int` and `epi_int_lite` N-body integrators (J. M. Hahn & J. N. Spitale 2013; J. M. Hahn et al. 2025) model a planetary ring as a set of gravitating streamlines whose particles are confined to the ring plane. Spiral bending waves, which are launched at a satellite's vertical resonances (F. H. Shu et al. 1983; F. H. Shu 1984), require the particles' vertical motions too. This paper describes `epi_int_3d`, a modest extension of `epi_int_lite` into the vertical (out-of-plane) direction, and uses it to simulate spiral bending waves.

2. VERTICAL EPICYCLIC MOTION

The `epi_int_lite` drift step advances each particle's unperturbed orbit about an oblate planet using first-order epicyclic equations that are exactly reversible: converting spatial coordinates into epicyclic orbit elements and back introduces no error, so the drift step is not a source of numerical drift (J. M. Hahn et al. 2025). Here those equations are extended to particles having vertical displacement z and vertical velocity v_z, with the same first-order accuracy and the same exact reversibility.

2.1. *Equations of motion*

A particle orbiting an oblate planet obeys Newton's second law,

$$\ddot{\mathbf{r}} = -\nabla\Phi, \tag{1}$$

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where Φ is the planet's gravitational potential. That potential is a series in the planet's zonal harmonics J_{2k} (Eqn. 5 of N. Borderies-Rappaport & P.-Y. Longaretti 1994), and as in J. M. Hahn & J. N. Spitale (2013) and J. M. Hahn et al. (2025) only the J_2 term is kept, so

$$\Phi(r,z) = -\frac{GM_p}{\rho}\left[1 - J_2\left(\frac{R_p}{\rho}\right)^2 P_2\left(\frac{z}{\rho}\right)\right]. \tag{2}$$

Here M_p is the planet's mass, R_p its radius, r the particle's cylindrical radius, $\rho^2 = r^2 + z^2$, and $P_2(\mu) = (3\mu^2 - 1)/2$. In cylindrical coordinates (r, θ, z), Eqn. (1) is

$$\begin{aligned} \ddot{r} - r\dot{\theta}^2 &= -\frac{\partial\Phi}{\partial r} \\ \frac{d}{dt}\left(r^2\dot{\theta}\right) &= -\frac{\partial\Phi}{\partial\theta} = 0 \\ \ddot{z} &= -\frac{\partial\Phi}{\partial z}. \end{aligned} \tag{3}$$

The potential is independent of θ, so the z-component of the specific angular momentum, $h = r^2\dot{\theta} = rv_\theta$, is conserved exactly, even when $z \neq 0$. The potential is also even in z, so near the ring plane

$$\Phi(r,z) = \Phi(r,0) + \frac{1}{2}\nu^2(r)z^2 + \mathcal{O}(z^4), \tag{4}$$

where $\Phi(r,0) = -GM_p/r - J_2GM_pR_p^2/2r^3$ is the midplane potential and

$$\nu^2(r) = \left.\frac{\partial^2\Phi}{\partial z^2}\right|_{z=0} = \frac{GM_p}{r^3}\left[1 + \frac{9}{2}J_2\left(\frac{R_p}{r}\right)^2\right]. \tag{5}$$

Equations (3) then become

$$\ddot{r} = \frac{h^2}{r^3} - \frac{\partial \Phi(r,0)}{\partial r} + \mathcal{O}(z^2)$$
$$\ddot{z} = -\nu^2(r) z + \mathcal{O}(z^3). \tag{6}$$

To first order the radial equation does not involve z; it is the planar equation of motion. J. M. Hahn et al. (2025) solved the radial part of Eqns. (6) to first order in small displacements from circular motion, in a form that is exactly reversible: the conversion from the coordinates r, θ, v_r, v_θ to the epicyclic elements $a, e, \tilde{\omega}, M$, and back, is exact. This is required if the solution is to be used in the drift step of a drift-kick N-body integrator. That solution is summarized in Appendix A. In it the particle's radius is $r = a[1 + \mathcal{O}(e)]$, its mean longitude $\lambda = \tilde{\omega} + (\Omega_a/\kappa_a)M$ advances at the mean angular velocity Ω_a, and its mean anomaly M advances at the epicyclic frequency κ_a. Note that Ω denotes the longitude of ascending node, while Ω_a is the mean angular velocity.

So the only additional physics that `epi_int_lite`'s drift step needs in order to follow out-of-plane motion is the vertical equation of motion, the second of Eqns. (6); the vertical forces that the ring and satellites exert enter the kick step and are treated separately. Evaluating $\nu^2(r)$ at $r = a$ reduces the vertical equation to the harmonic oscillator

$$\ddot{z} = -\nu_a^2 z, \tag{7}$$

with vertical frequency (Eqn. 17 of N. Borderies-Rappaport & P.-Y. Longaretti 1994)

$$\nu_a \equiv \nu(a) = \sqrt{\frac{GM_p}{a^3}} \left[1 + \frac{9}{2} J_2 \left(\frac{R_p}{a}\right)^2\right]^{1/2}, \tag{8}$$

where the neglected terms are of order $\mathcal{O}(ei)$, due to the particle's radial excursion $r - a = \mathcal{O}(ea)$, and $\mathcal{O}(i^3)$. Outside the planet the potential obeys Laplace's equation, which for an axisymmetric potential is

$$\nabla^2 \Phi = \frac{1}{r}\frac{\partial}{\partial r}\left(r \frac{\partial \Phi}{\partial r}\right) + \frac{\partial^2 \Phi}{\partial z^2} = 0. \tag{9}$$

At $r = a$ and $z = 0$, where $\Omega_a^2 = r^{-1}\partial\Phi/\partial r$, $\kappa_a^2 = 3\Omega_a^2 + \partial^2\Phi/\partial r^2$ and $\nu_a^2 = \partial^2\Phi/\partial z^2$, Eqn. (9) relates the three frequencies via $\kappa_a^2 + \nu_a^2 = 2\Omega_a^2$.

2.2. *Vertical solution*

The solution of Eqn. (7) is a sinusoid of frequency ν_a whose amplitude and phase define two new orbit elements, the inclination i and the longitude of ascending node Ω:

$$z = a\, i \sin(\lambda - \Omega)$$
$$v_z = a \nu_a i \cos(\lambda - \Omega), \tag{10}$$

where the node regresses at the rate

$$\dot{\Omega} = \Omega_a - \nu_a \simeq -\frac{3}{2} J_2 \left(\frac{R_p}{a}\right)^2 \Omega_a. \tag{11}$$

The vertical phase $\lambda - \Omega$ then advances at exactly ν_a, so Eqns. (10) solve Eqn. (7) exactly, with $v_z = \dot{z}$, and the particle crosses the ring plane upward at longitude $\lambda = \Omega$. Measuring the phase from λ rather than θ is what keeps the vertical motion a pure harmonic oscillator; $\theta - \Omega$ would add an $\mathcal{O}(ei)$ error to the phase.

2.3. *Coordinates to elements*

Equations (10) are inverted by a fifth step that follows steps 1–4 of Appendix A. Step 1 remains exact in three dimensions because h is conserved and r is the cylindrical radius, so steps 1–4 provide a, Ω_a, κ_a, e, M, $\tilde{\omega}$, and so λ. Step 5 computes ν_a from Eqn. (8) and then

$$i = \sqrt{\left(\frac{z}{a}\right)^2 + \left(\frac{v_z}{a\nu_a}\right)^2}$$
$$\Omega = \lambda - \mathrm{atan2}(z, v_z/\nu_a). \tag{12}$$

Once a is known, the mapping $(i, \lambda - \Omega) \leftrightarrow (z, v_z/\nu_a)$ is a change between polar and Cartesian coordinates, so Eqns. (10) and (12) are exact inverses. The drift step from t to $t + \Delta t$ converts coordinates to elements, advances $M \to M + \kappa_a \Delta t$ and $\Omega \to \Omega + (\Omega_a - \nu_a)\Delta t$, and converts back. Its vertical part is simply a rotation of $(z, v_z/\nu_a)$ through the angle $\nu_a \Delta t$.

3. THE GRAVITY OF A STREAMLINE

`Epi_int_3d` assigns particles to streamlines that are a closed eccentric path around the planet whose particles share a common semimajor axis a. All `epi_int_3d` simulations span only a narrow annulus within a broader planetary ring, so every particle is radially close to the nearby portions of all perturbing streamlines. Which is why `epi_int_3d` and its predecessors (`epi_int` and `epi_int_lite`) assume that the gravitational acceleration of a particle by the particles inhabiting the perturbing streamline at semimajor axis a' is that due to a straight and infinitely long wire:

$$\mathbf{A}_g = \frac{2G\lambda'}{\Delta} \hat{\mathbf{\Delta}}, \tag{13}$$

where λ' is the perturbing streamline's linear density, Δ is the particle's distance from the streamline's nearest point, which is labeled the near point in Fig. 1, and $\hat{\mathbf{\Delta}} = \mathbf{\Delta}/\Delta$ is the unit vector that points from the particle toward the near point.

3.1. *The perturbing streamline's geometry*

The following 3D treatment is an extension of the 2D force calculations used in J. M. Hahn & J. N. Spitale (2013). In cylindrical coordinates the particle is at $\mathbf{r} = (r, \theta, z)$. The ring matter that is represented by the particles in perturbing streamline a' is at $\mathbf{r}' = (r', \theta', z')$ that traces a continuous wire, so these coordinates are to be regarded as functions of longitude θ', i.e., $\mathbf{r}' = (r'(\theta'), \theta', z'(\theta'))$. Lagrange polynomials will be fit to the particles in streamline a', which means that $r'(\theta')$, $z'(\theta')$ and the streamline's linear density $\lambda'(\theta')$ can be regarded as known functions in the following. And because the particle is close to the streamline, $|r' - r| \ll r$, $|z' - z| \ll r$ and $|\theta' - \theta| \ll 1$, so the particle's local frame $(\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}}, \hat{\mathbf{z}})$ can be treated as Cartesian.

The vector from the particle to the streamline's near point (the black square in Fig. 1) can be written

$$\boldsymbol{\Delta} = \mathbf{r}'(\theta') - \mathbf{r} = \mathbf{d}_0 + \mathbf{d}_1, \tag{14}$$

where $\theta' = \theta + \delta\theta'$ is the near point's longitude as measured in the $z = 0$ plane, and $\delta\theta'$ is its longitude relative to the particle's with $|\delta\theta'| \ll 1$. This splits the trip to the near point across two legs. The first leg, $\mathbf{d}_0 = \mathbf{r}'(\theta) - \mathbf{r}$, travels from the particle to the crossing point $\mathbf{r}'(\theta) = (r'(\theta), \theta, z'(\theta))$, which is where the streamline crosses the particle's longitude (the cross in Fig. 1), so

$$\mathbf{d}_0 = d_r \hat{\mathbf{r}} + d_z \hat{\mathbf{z}}, \tag{15}$$

where $d_r = r'(\theta) - r$ and $d_z = z'(\theta) - z$. The second leg, $\mathbf{d}_1 = \boldsymbol{\Delta} - \mathbf{d}_0 = \mathbf{r}'(\theta') - \mathbf{r}'(\theta)$, then runs along the streamline from the crossing point to the near point. This is a right triangle (Fig. 1), so

$$|\mathbf{d}_0|^2 = \Delta^2 + |\mathbf{d}_1|^2. \tag{16}$$

That stretch of streamline is short and locally straight, so $\mathbf{d}_1 \simeq [r'(\theta') - r'(\theta)]\hat{\mathbf{r}} + r'(\theta)\,\delta\theta'\,\hat{\boldsymbol{\theta}} + [z'(\theta') - z'(\theta)]\hat{\mathbf{z}}$, which to first order in $\delta\theta'$ is

$$\mathbf{d}_1 \simeq (s_r \hat{\mathbf{r}} + \hat{\boldsymbol{\theta}} + s_z \hat{\mathbf{z}})\, r'\, \delta\theta' = N r'\, \delta\theta'\, \hat{\mathbf{t}}, \tag{17}$$

where r' is short for $r'(\theta)$. This uses $r'(\theta') - r'(\theta) \simeq (dr'/d\theta')\delta\theta'$ and likewise for z', where the streamline's radial and vertical slopes are

$$s_r = \frac{1}{r'}\frac{dr'}{d\theta'}\bigg|_{\theta'=\theta}, \qquad s_z = \frac{1}{r'}\frac{dz'}{d\theta'}\bigg|_{\theta'=\theta}, \tag{18}$$

which are evaluated at the perturbed particle's longitude $\theta' = \theta$ and are obtained from the derivatives of the Lagrange polynomials. Together they give $s = (s_r^2 + s_z^2)^{1/2}$, which is the streamline's slope relative to circular motion in the ring plane, i.e., the $\hat{\boldsymbol{\theta}}$ direction, so the streamline is tilted from $\hat{\boldsymbol{\theta}}$ by the angle φ, where $\tan\varphi = s$. Note that as $\delta\theta' \to 0$ the chord $\mathbf{d}_1$ points along the streamline, so $\hat{\mathbf{t}}$ is also the perturbing streamline's unit tangent:

$$\hat{\mathbf{t}} = \frac{s_r \hat{\mathbf{r}} + \hat{\boldsymbol{\theta}} + s_z \hat{\mathbf{z}}}{N}, \tag{19}$$

where $N^2 = 1 + s_r^2 + s_z^2$.

The near point is where $\boldsymbol{\Delta}$ is perpendicular to the streamline, $\boldsymbol{\Delta} \cdot \hat{\mathbf{t}} = (\mathbf{d}_0 + \mathbf{d}_1) \cdot \hat{\mathbf{t}} = 0$. And because $\mathbf{d}_1$ lies along $\hat{\mathbf{t}}$, this requires $\mathbf{d}_1 = -(\mathbf{d}_0 \cdot \hat{\mathbf{t}})\hat{\mathbf{t}}$, so

$$\boldsymbol{\Delta} = \mathbf{d}_0 + \mathbf{d}_1 = \mathbf{d}_0 - (\mathbf{d}_0 \cdot \hat{\mathbf{t}})\hat{\mathbf{t}}, \tag{20}$$

which is simply $\mathbf{d}_0$ with its component along the streamline removed.

Now equate $\mathbf{d}_1 = -(\mathbf{d}_0 \cdot \hat{\mathbf{t}})\hat{\mathbf{t}}$ to Eqn. (17), which shows that the near point lies at $\delta\theta' = -c/r'$, where $c = (\mathbf{d}_0 \cdot \hat{\mathbf{t}})/N = (d_r s_r + d_z s_z)/N^2$. So at the near point

$$\mathbf{d}_1 = -Nc\,\hat{\mathbf{t}} = -c(s_r \hat{\mathbf{r}} + \hat{\boldsymbol{\theta}} + s_z \hat{\mathbf{z}}). \tag{21}$$

Adding $\mathbf{d}_0$ to the above then yields $\boldsymbol{\Delta}$:

$$\begin{aligned} \boldsymbol{\Delta} &= (d_r - c s_r)\hat{\mathbf{r}} - c\hat{\boldsymbol{\theta}} + (d_z - c s_z)\hat{\mathbf{z}} \\ \Delta^2 &= d_r^2 + d_z^2 - (Nc)^2. \end{aligned} \tag{22}$$

Inserting $\boldsymbol{\Delta}$ into Eqn. (13) then provides the acceleration that streamline a' exerts on the particle:

$$\begin{aligned} A_{g,r} &= \frac{2G\lambda'(d_r - c s_r)}{\Delta^2} \\ A_{g,\theta} &= -\frac{2G\lambda' c}{\Delta^2} \\ A_{g,z} &= \frac{2G\lambda'(d_z - c s_z)}{\Delta^2}, \end{aligned} \tag{23}$$

where $N^2 = 1 + s_r^2 + s_z^2$ and $c = (d_r s_r + d_z s_z)/N^2$.

Lastly, note that because the streamline is tilted from $\hat{\boldsymbol{\theta}}$ by the angle φ, the right triangle of Eqn. (16) has $|\mathbf{d}_1| \le \Delta \tan\varphi = s\Delta$, with equality in the ring plane (Fig. 1). To lowest order in e and i, any disturbance of the ring is a superposition of modes whose streamlines are

$$\begin{aligned} r'(\theta) &= a'\,[1 - e\cos(\psi + \alpha)] \\ z'(\theta) &= a' i \sin(\psi + \beta), \end{aligned} \tag{24}$$

where $\psi = m(\theta - \Omega_p t)$, m is the mode's number of arms, Ω_p is its pattern speed, e and i are the eccentricity and inclination of the particles in streamline a', and α and β are phases. Equations (18) then give $s_r \simeq me\sin(\psi + \alpha)$ and $s_z \simeq mi\cos(\psi + \beta)$, so $s \lesssim m(e^2 + i^2)^{1/2}$, and the slopes of several modes simply add. Rings are nearly circular and nearly planar, and m is modest for the waves of interest, so $s \ll 1$. Consequently, $\varphi \ll 1$ and $|\mathbf{d}_1| \ll \Delta$, which means that the near point lies very close to the crossing point.

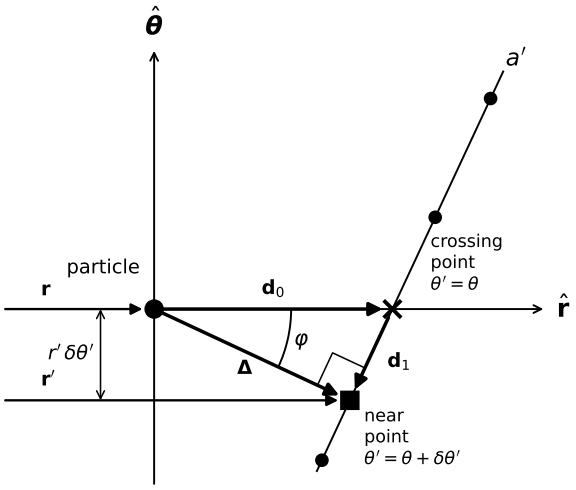


Figure 1. The gravity of a perturbing streamline a' on a particle in the ring plane: Fig. 3 of J. M. Hahn & J. N. Spitale (2013), redrawn. Near the particle's longitude, a' (thin line, with the three particles nearest in longitude) is straight, tilted from $\hat{\boldsymbol{\theta}}$ by φ, where $\tan\varphi = s_r$, and it crosses that longitude at $\mathbf{d}_0 = d_r\hat{\mathbf{r}}$ from the particle (cross). Its near point (black square) lies $\mathbf{d}_1$ further along it, $r'\,\delta\theta'$ along $\hat{\boldsymbol{\theta}}$ from the crossing point (here $\delta\theta' < 0$). The perpendicular $\boldsymbol{\Delta} = \mathbf{r}' - \mathbf{r} = \mathbf{d}_0 + \mathbf{d}_1$ (Eqn. 14), where $\mathbf{r}$ and $\mathbf{r}'$ are the positions of the particle and the near point (the planet is far off to the left), meets the streamline at a right angle, has length $\Delta = d_r\cos\varphi$ and makes the angle φ with $\hat{\mathbf{r}}$. The wire's acceleration $\mathbf{A}_g = (2G\lambda'/\Delta)\hat{\boldsymbol{\Delta}}$ points along it (Eqns. B3).

3.2. *Computing the acceleration*

The particles' coordinates are stored in arrays whose rows are the streamlines, and every timestep, before the accelerations are computed, the particles in each row are sorted by longitude. Because the streamlines are nearly concentric, the particle in row a' that has the perturbed particle's column index lies at nearly the same longitude, so it and its neighbor on either side bracket θ. Lagrange polynomials fit to those three particles then provide, by interpolation to θ, the crossing point's $r'(\theta)$ and $z'(\theta)$ and hence d_r and d_z, the linear density $\lambda'(\theta)$, and, from the polynomials' derivatives, the slopes s_r and s_z of Eqn. (18), and Eqns. (22)–(23) then provide the acceleration. This is repeated for every perturbing streamline, and the sum is the particle's acceleration due to ring gravity. Note that λ' is evaluated at the crossing point rather than at the near point, which is justified since Eqn. (13) assumes that λ' varies insignificantly over distances of several Δ along the streamline, while the near point lies only $|\mathbf{d}_1| \ll \Delta$ from the crossing point.

4. SPIRAL BENDING WAVES AT A VERTICAL RESONANCE

4.1. *Theory*

A satellite of mass M_s whose circular orbit of radius a_s is inclined by i_s to the ring plane launches spiral bending waves at its vertical resonances (F. H. Shu et al. 1983; F. H. Shu 1984). `Epi_int_3d` keeps only the resonant term of the satellite's potential, which accelerates a ring particle at longitude θ vertically by

$$A_z = -A_V\sin(m\theta - (m+1)\lambda_s + \Omega_s), \tag{25}$$

where

$$A_V = \frac{GM_s i_s}{2a_s^2} b_{3/2}^{(m)}(\beta) \tag{26}$$

is Eqn. (A21) of F. H. Shu (1984), $b_{3/2}^{(m)}$ is a Laplace coefficient evaluated at $\beta = r_V/a_s$, and the satellite's mean longitude $\lambda_s = \Omega_{a_s}t + \lambda_{s0}$ and longitude of ascending node $\Omega_s = (\Omega_{a_s} - \nu_{a_s})t + \Omega_{s0}$ advance at constant rates. The forcing's argument is then $\chi = m\theta - (m+1)\lambda_s + \Omega_s = m(\theta - \Omega_p t) + \chi_0$, which is an m-armed pattern that rotates with angular velocity $\Omega_p = \Omega_{a_s} + \nu_{a_s}/m$, where $\chi_0 = \Omega_{s0} - (m+1)\lambda_{s0}$. A particle feels the forcing at the frequency $\omega_D = m(\Omega_a - \Omega_p)$, and the inner vertical resonance lies at the radius r_V where $\omega_D = \nu_a$, which is $r_V = a_s[(m-1)/(m+1)]^{2/3}$ when the planet's oblateness is negligible.

Near r_V the ring's vertical displacement is $Z = \mathrm{Re}[h(x)e^{i\chi}]$, where $x = (r - r_V)/r_V$ is the fractional distance from resonance. Far from resonance, the complex amplitude h is, from Eqns. (88) and (91)–(92) of F. H. Shu (1984),

$$h = h_{\rm forced} + h_{\rm wave}, \quad h_{\rm forced} = \frac{L_V}{x},$$
$$h_{\rm wave} = \begin{cases} Z_w e^{-ix^2/2\epsilon} & x \ll -\epsilon^{1/2} \\ 0 & x \gg \epsilon^{1/2}, \end{cases} \tag{27}$$

where

$$\epsilon = \frac{2\pi G\sigma}{r_V\mathcal{D}}, \quad L_V = \frac{iA_V}{\mathcal{D}}, \tag{28}$$

σ is the ring's surface density, and $\mathcal{D} = r\,d(\nu_a^2 - \omega_D^2)/dr$ at r_V, which is $3(m-1)\Omega_a^2$ when oblateness is negligible.[2] The forced oscillation $h_{\rm forced}$ is the response of a massless ring, and $h_{\rm wave}$ is a trailing spiral bending wave on the inner side of the resonance that propagates inward. The wave's amplitude

$$|Z_w| = \left(\frac{2\pi}{\epsilon}\right)^{1/2}|L_V| = A_V\left(\frac{r_V}{G\sigma\mathcal{D}}\right)^{1/2} \tag{29}$$

[2] F. H. Shu (1984) writes perturbations as $e^{i(\omega t - m\theta)}$, the complex conjugate of the form used here, which reverses the sign of i in Eqns. (27)–(28).

is constant, while its wavelength

$$\lambda = \frac{4\pi^2 G\sigma}{\mathcal{D}|x|} \tag{30}$$

shrinks with distance from resonance.

Equations (27)–(30) give us a few opportunities to test `epi_int_3d`'s vertical motion. The forced oscillation alone is tested by a ring of massless particles (Section 4.2), which checks the forcing A_V, the resonance's location r_V and $\mathcal{D}$, which together also set the wave's amplitude. The wave is tested by a self-gravitating ring (Section 4.3). A third test may include the ring's viscosity, which damps the wave (Eqns. (97)–(98) of F. H. Shu 1984).

4.2. *Massless particles*

A ring of massless particles has no wave, only the forced oscillation $h_{\rm forced}$ of Eqn. (27), which, expressed as an inclination $|h_{\rm forced}|/r_V$, is

$$i_f = \frac{M_s}{M_p}\frac{\beta^2 b^{(m)}_{3/2}(\beta)}{6(m-1)|x|}\, i_s \tag{31}$$

when the planet's oblateness is negligible. The simulation uses $J_2 = 0.01$ and $R_p = 0.5$, in natural units where $G = M_p = 1$. Its ring is 81 streamlines of 6 particles each, spanning $1 \le r \le 1.04$, and an $m = 2$ satellite with $i_s = 0.01$ at $a_s = 2.125$ puts $r_V = 1.02$ mid-ring. The satellite's mass grows as $M_s = 10^{-6}(1 - e^{-t/\tau})$ with $\tau = 10^4$, or about 1600 orbits, slowly enough that the particles' motion is mostly forced rather than free, and what free motion remains averages away over time. Figure 2 shows that the particles' inclinations, averaged over the last 10^4 of the simulation's 2×10^4 orbits, are in excellent agreement with Eqn. (31), to within 4%, and the differences are of order x, which Eqn. (31) neglects. The test's notebook[3] also compares each particle's vertical motion over time to the exact solution, which the simulation follows to within 4×10^{-11} in z.

4.3. *A self-gravitating ring*

The same kind of satellite also launches a spiral bending wave in a ring that has mass. This ring is 540 streamlines of 32 particles each, spanning $1 \le r \le 1.008$, whose total mass 4×10^{-9} gives it the uniform surface density $\sigma = 7.9 \times 10^{-8}$. It has no viscosity or pressure, and J_2 and R_p are as in Section 4.2. An $m = 2$ satellite of mass $M_s = 2.5 \times 10^{-7}$ and inclination $i_s = 0.01$ at $a_s = 2.097$ puts the resonance at $r_V = 1.0064$, 80% of the way across the ring, and the satellite is at full mass from $t = 0$. These parameters do not correspond to any real ring and satellite. Rather, they were chosen so that the wave is linear and well resolved: its amplitude $|Z_w|$, Eqn. (29), keeps $|kZ_w| \le 0.03$ everywhere in the ring, and its wavelength, Eqn. (30), spans about 11 streamlines even at the ring's inner edge, where it is shortest.

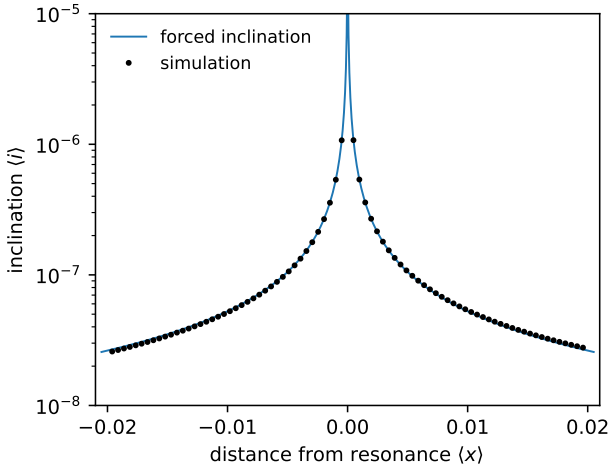


Figure 2. Inclinations $\langle i \rangle$ of massless particles near a satellite's $m = 2$ inner vertical resonance, averaged over each streamline's particles and over the last 10^4 orbits of the simulation of Section 4.2 (dots), plotted vs. their average fractional distance from resonance $\langle x \rangle$. The curve is the forced inclination, Eqn. (31). The streamline at $x = 0$, whose inclination grows without bound, is off scale.

The wave propagates inward at the group velocity $c_g = \pi G\sigma/\nu_a$ that follows from the bending-wave dispersion relation of Appendix D, so it takes the time $\tau_c = (r_V - 1)/c_g$, about 4100 orbits, to cross from r_V to the ring's inner edge. The simulation stops at 3500 orbits, or $0.86\,\tau_c$, before the wave can reflect there. Figure 3 shows the wave being launched at r_V and propagating inward. Switching the satellite on at full mass also starts a free vertical oscillation everywhere, so at first every streamline moves as a massless particle would. The ring's self-gravity then takes over: the response near r_V settles onto linear theory, and the wave front advances inward at about c_g. The front overshoots linear theory a little, since it carries along the free oscillation that was started at resonance, and a small residue of that oscillation remains ahead of the front. Behind the front, though, the simulated inclinations are in excellent agreement with linear theory, with about 14 wavelengths of the wave in place by the end of the run. There the wave's amplitude is within 3% of $|Z_w|$, and its phase winds inward as the trailing wave of Eqn. (27) does. On the nonwave side of the resonance, the forced response

[3] https://github.com/joehahn/epi_int_3d/blob/main/tests/vertical_resonance/vertical_resonance.ipynb

is up to 9% larger than linear theory's. That theory is for an infinite sheet, which has ring matter beyond this ring's outer edge, and that is the likely cause, though it has not been quantified. The satellite pulls only vertically, and the wave leaves no mark in the ring plane: no particle moves radially by more than 4×10^{-8}, well under 1% of the streamline spacing $\Delta = 1.5 \times 10^{-5}$.

The simulated wave's wavenumber, measured from the radial gradient of its phase, is 5.5% larger than Eqn. (30)'s (the median behind the front), i.e., its wavelength is that much shorter. This is because the simulated ring is a finite lattice of streamlines rather than the infinite, continuous sheet that Eqn. (30) assumes, and both differences weaken the wave's restoring force. Appendix D shows that the lattice's spacing reduces self-gravity's restoring force from $2\pi G\sigma|k|$ to $X(k\Delta)$, and that the ring's edges reduce it further, by $2G\sigma(1/d_{\rm in} + 1/d_{\rm out})$. With the lattice allowed for, the simulated wavenumber is 2.3% larger than expected, and with the edges allowed for too, Eqn. (D12), it is 0.9% larger. The remainder is the lattice's: in a ring half as wide, doubling the number of streamlines lowers the wavenumber's excess over Eqn. (30) from 7.2% to 4.8%, while its excess over Eqn. (D12) is then 0.5%. The test's notebook[4] also compares the wave's complex amplitude and the satellite's torque on the ring with linear theory.

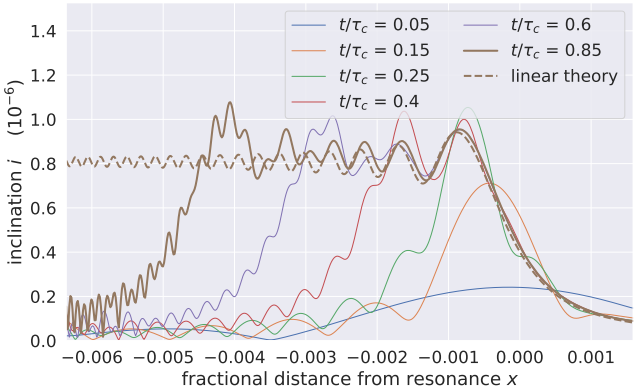


Figure 3. Inclinations i of the streamlines in the self-gravitating ring of Section 4.3, averaged over each streamline's particles, plotted vs. their fractional distance from resonance x, at six times t in units of the time τ_c that the wave takes to propagate from r_V to the ring's inner edge, which is at the left. The dashed curve is linear theory's inclination $|h|/r$, where h is the trailing solution of Eqn. (88) of F. H. Shu (1984), which tends to Eqns. (27) away from resonance.

APPENDIX

A. THE IN-PLANE SOLUTION OF HAHN ET AL. (2025)

This appendix summarizes the first-order, exactly reversible solution of the planar equation of motion from Appendix A of J. M. Hahn et al. (2025), in the notation used here. The mean angular velocity and epicyclic frequency at semimajor axis a are

$$\Omega_a = \sqrt{\frac{GM_p}{a^3}}\left[1 + \frac{3}{2}J_2\left(\frac{R_p}{a}\right)^2\right]^{1/2}$$
$$\kappa_a = \sqrt{\frac{GM_p}{a^3}}\left[1 - \frac{3}{2}J_2\left(\frac{R_p}{a}\right)^2\right]^{1/2}. \tag{A1}$$

In terms of the epicyclic elements a, e, $\tilde{\omega}$ and M, the particle's coordinates are

$$r = \frac{a}{\sqrt{1 + 2e\cos M}}$$
$$\theta = \frac{\Omega_a}{\kappa_a}(M + 2e\sin M) + \tilde{\omega}$$
$$v_r = ea\kappa_a \sin M \left(\frac{r}{a}\right)^3$$
$$v_\theta = \frac{a^2\Omega_a}{r}, \tag{A2}$$

which conserve $h = rv_\theta = a^2\Omega_a$ exactly and solve the planar equation of motion with errors of order $\mathcal{O}(e^2)$. The elements advance as $M \to M + \kappa_a\Delta t$, with a, e and $\tilde{\omega}$ constant. Equations (A2) are inverted exactly by:

[4] https://github.com/joehahn/epi_int_3d/blob/main/tests/bending_waves/bending_waves.ipynb

1. Compute $h = rv_\theta$, then solve $h^2 = a^4\Omega_a^2$ for the semimajor axis, $a = R_p\left(C + \sqrt{C^2 - 3J_2/2}\right)$ where $C = h^2/2GM_pR_p$.
2. Compute Ω_a and κ_a from Eqns. (A1).
3. Compute $e\sin M = (v_r/a\kappa_a)(a/r)^3$ and $e\cos M = \frac{1}{2}[(a/r)^2 - 1]$, which give e and M.
4. Compute $\tilde{\omega} = \theta - (\Omega_a/\kappa_a)(M + 2e\sin M)$.

B. RELATION TO EPI_INT AND EPI_INT_LITE

In the ring plane $d_z = s_z = 0$, and the acceleration of Section 3.1 reduces to the geometry of Fig. 3 of J. M. Hahn & J. N. Spitale (2013), which is redrawn here as Fig. 1. There the streamline's tilt φ from $\hat{\boldsymbol{\theta}}$ has $\tan\varphi = s_r$. Then $c = d_r\sin\varphi\cos\varphi$ and $\boldsymbol{\Delta} = d_r\cos\varphi(\cos\varphi\hat{\mathbf{r}} - \sin\varphi\hat{\boldsymbol{\theta}})$, i.e., the perpendicular has length $\Delta = d_r\cos\varphi$ and makes the angle φ with $\hat{\mathbf{r}}$. The acceleration is then

$$\begin{aligned} A_{g,r} &= \frac{2G\lambda'}{d_r} \\ A_{g,\theta} &= -\frac{2G\lambda'\tan\varphi}{d_r}. \end{aligned} \tag{B3}$$

Out of the plane, the tangential acceleration $A_{g,\theta} = -2G\lambda' c/\Delta^2$ of Eqns. (23) also includes the term $-2G\lambda' d_z s_z/N^2\Delta^2$, which is due to the streamline's vertical tilt, and this is how vertically displaced streamlines exchange angular momentum.

J. M. Hahn & J. N. Spitale (2013) give the components of this acceleration as $A_{g,r} = A_g\cos\varphi$ and $A_{g,\theta} = -A_g\sin\varphi$, with $A_g = 2G\lambda/\Delta$ and Δ the perpendicular distance, and approximate them as $A_{g,r} \simeq A_g$ and $A_{g,\theta} \simeq -A_g v_r'/v_\theta'$ (their Eqns. 20), to lowest order in the streamline's eccentricity. With Δ read as the radial gap d_r and $\tan\varphi = v_r'/v_\theta'$, those are Eqns. (B3) without approximation. `Epi_int_lite` (J. M. Hahn et al. 2025) differs from Eqns. (B3) in three ways, which `epi_int_3d` corrects.

Distance—`Epi_int_lite` evaluates the pull's magnitude at the crossing point, $\Delta = d_r$, but keeps its direction toward the near point,

$$\begin{aligned} A_{g,r} &= \frac{2G\lambda\cos\varphi}{d_r} \\ A_{g,\theta} &= -\frac{2G\lambda\sin\varphi}{d_r}. \end{aligned} \tag{B4}$$

This pull is weaker than Eqns. (B3) by the factor $\cos\varphi = 1 - \mathcal{O}(\varphi^2)$. Which is why `epi_int` and `epi_int_lite` could use the distance to the crossing point, $|\mathbf{d}_0|$, for Δ, also out of the plane, i.e., neglect $\mathbf{d}_1$ in the pull's magnitude. Section 3.1 shows that the near point lies very close to the crossing point, $|\mathbf{d}_1| \le s\Delta$ with $s \lesssim m(e^2 + i^2)^{1/2}$, and inserting that into $|\mathbf{d}_0|^2 = \Delta^2 + |\mathbf{d}_1|^2$ (Eqn. 16) gives $|\mathbf{d}_0|/N \le \Delta \le |\mathbf{d}_0|$. So the two differ by a fraction of order $s^2 \lesssim m^2(e^2 + i^2)$, which is $\lesssim 10^{-4}$ even for $m(e^2 + i^2)^{1/2}$ as large as 10^{-2} and so of no practical consequence, but Eqns. (B3) cost no more.

Direction—`Epi_int_lite` takes the wire's direction from the streamline's velocity at the crossing point, $\sin\varphi = v_r'/v'$ and $\cos\varphi = v_\theta'/v'$ (Fig. 4). The velocity is tangent to the streamline only when the streamline's shape is stationary. If the shape is a pattern that rotates at Ω_p while its particles orbit at Ω_a, then $v_r'/v_\theta' = (1 - \Omega_p/\Omega_a)s_r$, where s_r is the slope of the streamline's shape (Eqn. 18). Near an m-armed Lindblad resonance $1 - \Omega_p/\Omega_a \simeq \pm 1/m$, so the velocity underestimates the wire's tilt, and with it the tangential pull that transports angular momentum between streamlines, by a factor of about m. For the slowly precessing $m = 1$ ringlets of J. M. Hahn et al. (2025), Ω_p is the precession rate and the error is negligible. `Epi_int_3d` takes the direction from the shape (Eqn. 19).

Linear density—`Epi_int_lite` evaluates every perturbing streamline's pull with the perturbed particle's own linear density λ, rather than the perturbing streamline's λ'. The difference, $\lambda' - \lambda \simeq (a' - a)\partial\lambda/\partial a$, adds a nearly uniform pull of order the ring's mass over the planet's; in the eccentric ringlets of J. M. Hahn et al. (2025) it changes the precession rate by about 10^{-7} of itself.

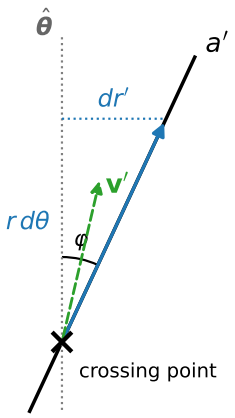


Figure 4. The perturbing streamline at its crossing point (Fig. 1). Its shape has the slope $\tan\varphi = s_r$: a step $r\,d\theta$ along $\hat{\boldsymbol{\theta}}$ moves it dr' outward. Its velocity $\mathbf{v}'$ (dashed), which `epi_int_lite` takes as the wire's direction, is tilted less when the shape rotates, here by half as much, as for an $m = 2$ pattern.

In an eccentric ringlet the three corrections together change the streamlines' eccentricities by less than 10^{-4} of themselves over 5000 orbits. Only the second matters for spiral waves, where it changes the torque between streamlines by a factor of about m. As in `epi_int_lite`, the particle's own streamline exerts no force on it in `epi_int_3d`, and Δ never vanishes, since streamlines may not cross. The slopes require that the streamline be well sampled by its particles. `Epi_int_3d` also keeps `epi_int_lite`'s ring gravity, Eqns. (B4), as an option, so that its results can be checked against `epi_int_lite`'s exactly.

C. RESOLUTION LIMIT OF THE STREAMLINE METHOD

A streamline model represents a continuous ring by N_r streamlines spaced Δ apart, and that discreteness sets which wavelengths of the ring's response it can follow. This appendix derives the axisymmetric stability of a uniform lattice of circular streamlines under the self-gravity and pressure accelerations of `epi_int_lite` (J. M. Hahn & J. N. Spitale 2013; J. M. Hahn et al. 2025), and compares it with the continuum Toomre criterion.

Each streamline of linear density λ exerts the acceleration $2G\lambda/d$ of a straight wire on a particle a distance d away (J. M. Hahn & J. N. Spitale 2013). The pressure $p = c^2\sigma$ acts through the surface density $\sigma_j = 2\lambda/(r_{j+1} - r_{j-1})$ of streamline j and the central difference $A_j = -(p_{j+1} - p_{j-1})/2\lambda$, which tends to $-\sigma^{-1}\partial p/\partial r$ as $\Delta \to 0$. Displace the streamlines radially by $\xi_j = \xi\cos(j\phi)$, where $\phi = k\Delta$ and k is the radial wavenumber, so that $0 < \phi \le \pi$; $\phi = \pi$ is the shortest wave the lattice can carry, the zigzag $\xi_j = (-1)^j\xi$ of Fig. 5a. Linearizing in ξ, the streamlines at $j = \pm n$ add $4G\lambda(1 - \cos n\phi)\xi/(n\Delta)^2$ to the acceleration of streamline 0, and summing over n with $\sigma = \lambda/\Delta$ gives

$$X(\phi) = \frac{4G\sigma}{\Delta}\sum_{n=1}^{\infty}\frac{1-\cos n\phi}{n^2} = \frac{4G\sigma}{\Delta}\left(\frac{\pi\phi}{2} - \frac{\phi^2}{4}\right), \tag{C5}$$

while pressure contributes

$$\omega_p^2(\phi) = \left(\frac{c}{\Delta}\right)^2\sin^2\phi, \tag{C6}$$

so that each mode oscillates at the frequency

$$\omega^2 = \kappa^2 - X(\phi) + \omega_p^2(\phi). \tag{C7}$$

For $\phi \ll 1$ these recover the continuum terms $X \to 2\pi G\sigma k$ and $\omega_p^2 \to c^2k^2$, and Eqn. (C7) becomes the dispersion relation of a pressure-supported self-gravitating sheet, which is stable at all k when $Q = c\kappa/\pi G\sigma \ge 1$ (A. Toomre 1964; J. Binney & S. Tremaine 1987). At the grid scale the lattice departs from the continuum in two ways (Fig. 5b). Self-gravity peaks at $X(\pi) = \pi^2 G\sigma/\Delta$, half the continuum value $2\pi G\sigma k$ at $k = \pi/\Delta$. And pressure vanishes there: in the zigzag, streamlines $j \pm 1$ move together, so $r_{j+1} - r_{j-1}$ and hence σ_j and p_j do not change. The viscous fluxes of J. M. Hahn & J. N. Spitale (2013), which are differenced in the same way, likewise exert no force on the zigzag.

A cold continuum disk ($c = 0$) is unstable to all wavelengths shorter than $\lambda_{\rm crit} = 4\pi^2 G\sigma/\kappa^2$, and the lattice hides these: by Eqn. (C5) a cold lattice is axisymmetrically stable as long as

$$\mathcal{R} \equiv \frac{\lambda_{\rm crit}}{2\Delta} = \frac{2\pi^2 G\sigma}{\kappa^2\Delta} \simeq \frac{\pi G m_r N_r}{\kappa^2 a \Delta a^2} < 2, \tag{C8}$$

where 2Δ is the shortest resolved wavelength, and the last form is for a ringlet of mass m_r, semimajor axis a and width Δa, with $\sigma \simeq m_r/2\pi a\Delta a$ and $\Delta \simeq \Delta a/N_r$. Adding streamlines to a given ringlet raises $\mathcal{R}$ in proportion to N_r.

A viscous ringlet must also avoid the viscous overstability, the growth of axisymmetric epicyclic oscillations that viscous stresses drive when the shear stress rises steeply enough with surface density (U. Schmit & W. M. Tscharnuter 1995; P.-Y. Longaretti & N. Rappaport 1995; H. Salo et al. 2001; J. Schmidt et al. 2001; H. N. Latter & G. I. Ogilvie 2006, 2008). For a cold ringlet with constant kinematic shear and bulk viscosities ν_s and ν_b, as in `epi_int_lite`, and Keplerian $\kappa = \Omega$, a mode with oscillation frequency $\omega \simeq \kappa(1-x)^{1/2}$ grows to first order in the viscosities at the rate

$$\gamma = \frac{K}{6(1-x)}\left[\nu_s(4x+2) - 3\nu_b(1-x)\right], \tag{C9}$$

where $K = k^2$ and $x = 2\pi G\sigma k/\kappa^2$ in the continuum. Without self-gravity ($x = 0$) the ringlet is stable when $\nu_b > 2\nu_s/3$, the onset found by U. Schmit & W. M. Tscharnuter (1995) and M. Lehmann et al. (2017) for a viscosity independent of σ, while self-gravity is destabilizing (M. Lehmann et al. 2017): a mode is stable only for $x < (3\nu_b - 2\nu_s)/(4\nu_s + 3\nu_b)$, which is 1/7 for $\nu_b = \nu_s$ and 0.4 for $\nu_b = 2\nu_s$. The streamline lattice follows Eqn. (C9) with the grid's versions of both terms: $K = \sin^2\phi/\Delta^2$, from the code's central differences of the viscous stresses, and $x = X(\phi)/\kappa^2$. Both stresses vanish at the zigzag, $\phi = \pi$, where self-gravity is strongest, $X(\pi)/\kappa^2 = \mathcal{R}/2$; so a viscous lattice is stable at every wavelength when $\mathcal{R} < 2/7$ for $\nu_b = \nu_s$, or $\mathcal{R} < 4/5$ for $\nu_b = 2\nu_s$, more stringent than Eqn. (C8). Since $K\nu \sim \nu/\Delta^2$, an overstable lattice's modes grow faster the more streamlines it has.

That limit matters because a ringlet's self-gravity must be strong enough to resist the differential precession due to J_2. Across a ringlet of width Δa, J_2 drives a differential precession $\propto J_2(R_p/a)^2\Omega_a\Delta a/a$, while for a small nonlinearity parameter the ringlet's self-gravity drives one $\propto (m_r/M_p)\Omega_a\Delta\tilde{\omega}(a/\Delta a)^2$, where $\Delta\tilde{\omega}$ is the periapse twist across it (cf. Eqn. C1 of J. M. Hahn et al. 2025). So keeping J_2 from winding up the ringlet requires $m_r \propto J_2\Delta a^3/\Delta\tilde{\omega}$, and then $\mathcal{R} \propto J_2\Delta a N_r/\Delta\tilde{\omega}$: a wider ringlet does not help, and for a given planet more streamlines can be afforded only by a narrower ringlet. This is tested here with simulations of a ringlet of width $\Delta a = 0.0125a$ whose edges are tapered, mass $m_r = 3\times10^{-8}M_p$, eccentricity $e = 10^{-4}$, periapses twisted by π from its inner to outer edge, and kinematic viscosity $\nu = 7\times10^{-11}a^2\Omega_a$, with no pressure, $R_p = 0.5a$ and $J_2 = 0.01$. With $N_r = 100$ streamlines ($\mathcal{R} \simeq 0.09$ in the ringlet's middle, where the taper makes σ 1.25 times and Δ 0.8 times their ringlet-wide means) the ringlet stays smooth for 10^4 orbits while self-gravity, rather than J_2, controls its shape: within about 1000 orbits it straightens the initial twist, and the ringlet's body then stays nearly aligned, as two sub-ringlets that precess slowly relative to each other. With $N_r = 300$ ($\mathcal{R} \simeq 0.3$) and $N_r = 1000$ ($\mathcal{R} \simeq 0.9$) it becomes ragged and its streamlines cross after about 190 and 30 orbits. At $N_r = 1000$ the disturbance starts in the ringlet's interior, grows at the same rate with 40 or 241 particles per streamline, and does not grow when ring gravity is turned off, consistent with the self-gravity-assisted overstability for $\nu_b = \nu_s$ at $\mathcal{R} > 2/7$. Pressure with $Q \le 2$ delays the crossing by less than two orbits, as expected from Fig. 5b, where $\omega_p^2 \le 0.1\kappa^2$.

The crossings near $\mathcal{R} \simeq 0.3$ are mostly a different, numerical, problem. The same ringlet without the initial twist, at $N_r = 333$ ($\mathcal{R} \simeq 0.31$), crosses after 145 orbits, and its code's own linearized axisymmetric modes, computed from the reference implementation's accelerations, grow by only 3.3×10^{-3} per orbit, far too slowly. The disturbance that crosses is instead a sawtooth along each streamline with azimuthal wavenumber $m \simeq N_t/2$, where N_t is the number of particles per streamline. It also grows in a circular ringlet, without ring gravity, and without viscosity, though more slowly (after about 220, 170 and 850 orbits), its onset varies non-monotonically with N_t (with $N_t = 30$ or 50 the ringlet survives 10^4 orbits, with 20, 40 or 320 it does not), and pressure with $Q \le 10$ delays it by at most 20%. It is presumably seeded by the 3-point Lagrange interpolation in longitude that couples neighboring streamlines' particles, and it remains an open problem.

So Eqn. (C8) is a necessary condition only. In practice a simulated ringlet whose self-gravity matters should be run with $\mathcal{R} \lesssim 0.1$, viscous or not, and a ringlet that needs more streamlines must be made narrower and much less massive.

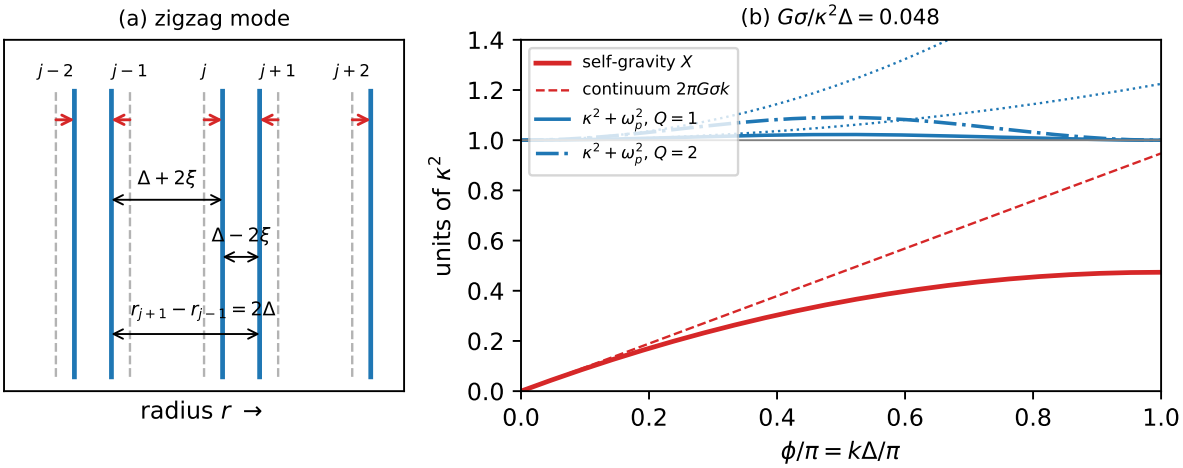


Figure 5. (a) The zigzag mode of a lattice of streamlines (blue) displaced by $\pm\xi$ from their unperturbed radii (dashed): the gaps alternate between $\Delta \pm 2\xi$, but $r_{j+1} - r_{j-1} = 2\Delta$ does not change, so neither does the surface density that sets streamline j's pressure. (b) The terms of Eqn. (C7) versus $\phi = k\Delta$, in units of κ^2, for $G\sigma/\kappa^2\Delta = 0.048$ (the ringlet of the text with $N_r = 1000$): the destabilizing self-gravity X (Eqn. C5, solid red) and its continuum value $2\pi G\sigma k$ (dashed red), and the restoring $\kappa^2 + \omega_p^2$ for $Q = 1$ and 2 (blue; Eqn. C6) and their continuum values $\kappa^2 + c^2k^2$ (dotted blue). A mode is unstable where X exceeds $\kappa^2 + \omega_p^2$.

D. THE RESTORING FORCE OF A BENDING WAVE

Equation (13) recovers the self-gravity of a bending wave. Consider a sheet of surface density σ made of streamlines with $\lambda = \sigma\, dr'$, displaced vertically to $Z(r) = Z_0 \cos kr$. When $|d_z| \ll |d_r|$ and the slopes are small, $A_{g,z} \simeq 2G\lambda d_z/d_r^2$, and the sum of $A_{g,z}$ over the sheet is

$$A_z = 2G\sigma Z_0 \int_{-\infty}^{\infty} \frac{\cos kx - 1}{x^2}\, dx = -2\pi G\sigma |k| Z, \tag{D10}$$

which is the self-gravity term of the bending-wave dispersion relation, $(\omega - m\Omega_a)^2 = \nu_a^2 + 2\pi G\sigma|k|$ (F. H. Shu et al. 1983; F. H. Shu 1984). So summing the wires' pulls over the streamlines gives the vertical restoring force of a self-gravitating sheet, just as the same sum gives the $-2\pi G\sigma|k|$ term of a density wave in the plane. For bending waves, though, self-gravity is restoring, and adds to ν_a^2.

On the code's lattice of streamlines spaced Δ apart (here Δ is the spacing, as in Appendix C), with $Z_j = Z_0 \cos(j\phi)$ and $\phi = k\Delta$,

$$A_z = \frac{4G\sigma Z_0}{\Delta} \sum_{n=1}^{\infty} \frac{\cos n\phi - 1}{n^2} = -X(\phi) Z_0, \tag{D11}$$

where $X(\phi)$ is the lattice self-gravity of Eqn. (C5). The vertical lattice modes therefore oscillate at $\omega^2 = \nu_a^2 + X(\phi)$, which tends to the continuum value as $\phi \to 0$. At the zigzag $\phi = \pi$, the extra restoring force is half the continuum value, as it is in the plane. Vertically, however, the lattice's self-gravity is stabilizing, so the resolution limit of Eqn. (C8), which concerns in-plane modes, has no vertical counterpart.

A ring of finite width also lacks the streamlines beyond its edges. Consider a streamline whose Nth neighbor on one side would be the first one beyond the ring's edge. The sum of Eqn. (D11) then stops at $n = N - 1$ on that side, and the missing terms are $(2G\sigma Z_0/\Delta) \sum_{n \geq N} (\cos n\phi - 1)/n^2$. Their cosines nearly cancel when the wave is shorter than the distance to the edge, while $\sum_{n \geq N} 1/n^2 \simeq 1/(N - \frac{1}{2})$, so the missing pull is about $2G\sigma Z_0/d$, where $d = (N - \frac{1}{2})\Delta$ is the streamline's distance to the ring's edge plus half a spacing. Near r_V, where $\omega_D^2 - \nu_a^2 \simeq \mathcal{D}|x|$ on the wave side, the lattice's restoring force so replaces the continuum's $\mathcal{D}|x| = 2\pi G\sigma|k|$ with

$$\mathcal{D}|x| = X(k\Delta) - 2G\sigma \left(\frac{1}{d_{\text{in}}} + \frac{1}{d_{\text{out}}} \right), \tag{D12}$$

where $d_{\rm in}$ and $d_{\rm out}$ are the distances to the ring's inner and outer edges, so measured. Both terms lower the restoring force at a given k, so the wave's wavenumber is larger than in the continuum, by the amounts that Section 4.3 finds.

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